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-- 1. DDL for actors table
create type film_role as
(
    filmid text,
    film    text,
    votes   integer,
    rating  real
);

create type quality_class as enum ('star', 'good', 'average', 'bad');

create table actors
(
    actor          text,
    actorid        text,
    films          film_role[]    not null,
    quality_class  quality_class not null,
    is_active      boolean,
    current_year   integer,
    primary key (actorid, current_year)
);

-- 2. Cumulative table generation query
DO
$$
DECLARE
    start_year integer := (select min(Year)
                           from actor_films);
    end_year   integer := (select max(Year)
                           from actor_films);
BEGIN
    FOR curr_year IN start_year..end_year
    LOOP
        insert into actors
        with yesterday as (select *
                           from actors
                           where current_year = curr_year - 1),
             today as (select ActorId,
                             min(Actor)           as Actor,
                             array_agg(row (
                                 FilmID,
                                 Film,
                                 votes,
                                 Rating
                             )::film_role) as films,
                             curr_year      as Year,
                             avg(Rating)    as Rating
                           from actor_films
                           where Year = curr_year
                           group by ActorId)
        select coalesce(y.actor, t.Actor)           as actor,
               coalesce(y.actorid, t.ActorId)       as actorid,
               case
                   when y.films is null
                   then t.films
                   when t.ActorId is not null

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order by current_year) as prev_is_active
        from actors
        where current_year <= curr_year),
    with_indicators as (select *,
        case
            when quality_class <>
prev_quality_class then 1
            when is_active <>
prev_is_active then 1
            else 0
            end as change_indicator
        from with_previous),
    with_streaks as (select *,
        sum(change_indicator)
        over (partition by actorid order
by current_year) as streak_identifier

        from with_indicators)
    select actorid,
        actor,
        quality_class,
        is_active,
        min(current_year) as start_year,
        max(current_year) as end_year,
        curr_year as current_year
    from with_streaks
    group by actorid, streak_identifier, actor, quality_class,
is_active
        order by actorid;
    END LOOP;
END
$$;

-- Get scd records number per year
select current_year, count(*) as num_records
from actors_history_scd
group by current_year
order by 1 desc;

-- Get scd records as of last_year
with params as (select max(current_year) as last_scd_year from
actors_history_scd)
select scd.*
from actors_history_scd scd
    join params p
        on scd.current_year = p.last_scd_year
order by actorid, start_year
;

-- Fill with new actors data as of 1982
insert into actors_history_scd
with params as (select 1982 as curr_year),
    scd_as_of_last_year as (select scd.*
        from actors_history_scd scd
            join params p
                on scd.current_year = p.curr_year
- 1),

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scd_ongoing_streaks_as_of_last_year as (select actorid,
                                             actor,
                                             quality_class,
                                             is_active,
                                             start_year,
                                             end_year
                                             from scd_as_of_last_year
                                             where end_year = current_year),

this_year_data as (select a.*
                      from actors a
                      join params p
                        on a.current_year = p.curr_year),

unchanged_scd as (select td.actorid,
                        td.actor,
                        td.quality_class,
                        td.is_active,
                        ls.start_year,
                        td.current_year
                      from this_year_data td
                      join scd_ongoing_streaks_as_of_last_year ls
                        on td.actorid = ls.actorid and
td.current_year = ls.end_year + 1
                        where td.quality_class = ls.quality_class
                        and td.is_active = ls.is_active),

changed_scd as (select td.actorid,
                      td.actor,
                      td.quality_class,
                      td.is_active,
                      td.current_year,
                      td.current_year
                    from this_year_data td
                    join scd_ongoing_streaks_as_of_last_year ls
                      on td.actorid = ls.actorid and
td.current_year = ls.end_year + 1
                      where td.quality_class <> ls.quality_class
                      or td.is_active <> ls.is_active),

history_scd as (select actorid,
                      actor,
                      quality_class,
                      is_active,
                      start_year,
                      end_year
                    from scd_as_of_last_year scd
                    where not exists(select 1
                                      from unchanged_scd uscd
                                      where scd.actorid = uscd.actorid
                                      and scd.start_year =
uscd.start_year)),

new_scd as (select td.actorid,
                  td.actor,
                  td.quality_class,
                  td.is_active,
                  td.current_year,
                  td.current_year
                from this_year_data td
                left join scd_ongoing_streaks_as_of_last_year ls
                  on td.actorid = ls.actorid and

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td.current_year = ls.end_year + 1
    where ls.actorid is null),
    accumulated_scd as (select *, 'history' as source
                        from history_scd
                        union all
                        select *, 'unchanged' as source
                        from unchanged_scd
                        union all
                        select *, 'changed' as source
                        from changed_scd
                        union all
                        select *, 'new' as source
                        from new_scd)

select acc.actorid,
       acc.actor,
       acc.quality_class,
       acc.is_active,
       acc.start_year,
       acc.end_year,
       p.curr_year as current_year
from accumulated_scd acc,
     params p
;

-- Get scd records number per year
select current_year, count(*) as num_records
from actors_history_scd
group by current_year
order by 1 desc;

-- Get scd records as of last_year
with params as (select max(current_year) as last_scd_year from
actors_history_scd)
select scd.*
from actors_history_scd scd
     join params p
       on scd.current_year = p.last_scd_year
order by actorid, start_year
;

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